

Quality Assessment & Improvement of the smoltcp TCP/IP Stack

Security Testing & Software Quality Assurance

smoltcp v0.12.0 — Open-Source Embedded Network Stack

Final Project Presentation | April 2026

Agenda

1 Introduction & Scope

2 Mutation Adequacy Testing

3 Input Space Partitioning

4 Security Fuzzing

5 Conformance Testing

6 Performance Testing

7 White-Box Coverage

8 Software Safety

9 Quality Dashboard

10 Conclusions

Introduction

What is smoltcp?

A standalone, event-driven TCP/IP stack written in **Rust**, targeting embedded, bare-metal, and resource-constrained systems.

- Rust's memory-safety eliminates buffer overflows, use-after-free, and double-free vulnerabilities
- Logic errors, spec deviations, and performance regressions remain possible
- Our assessment spans **8 quality dimensions**

Platforms: Windows 10 (MSVC) | Ubuntu 24.04.2 LTS (GNU/LLVM)

| Scope & Target Modules

Four highest-complexity modules:

- `storage/assembler.rs`
TCP segment reassembly
- `wire/ipv4.rs`
IPv4 header parsing
- `wire/udp.rs`
UDP datagram handling
- `wire/tcp.rs`
TCP segment parsing & emission

Eight quality dimensions:

1. Mutation adequacy
2. Test suite improvement
3. Input space partitioning
4. Security fuzzing
5. RFC conformance
6. Performance benchmarking
7. White-box coverage
8. Software safety

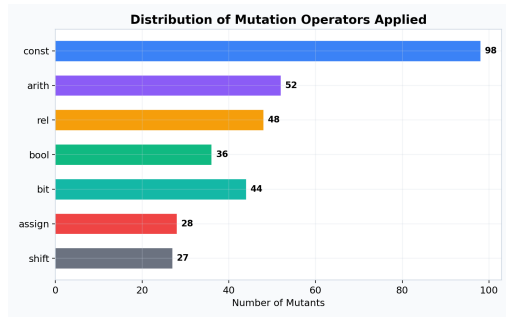
Custom Python-based mutation engine with 333 first-order mutants

Mutation Testing — Operator Distribution

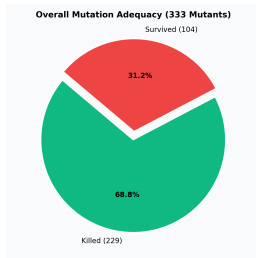
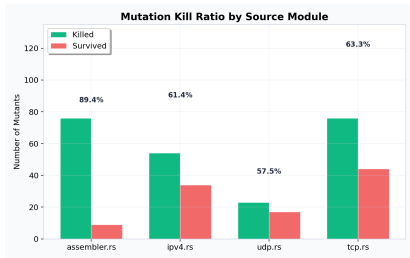
7 mutation operator classes:

- Constant replacement (98)
- Arithmetic operator (52)
- Relational operator (48)
- Boolean connector (36)
- Bitwise operator (44)
- Assignment operator (28)
- Shift operator (27)

333 total first-order mutants



Mutation Testing — Kill Ratios



Module	Total	Killed	Survived	Kill %
assembler.rs	85	76	9	89.41%
ipv4.rs	88	54	34	61.36%
udp.rs	40	23	17	57.50%
tcp.rs	120	76	44	63.33%
Overall	333	229	104	68.77%

Equivalent Mutant Analysis

- Manual review of **top 5 surviving mutants** (lowest-index survivors)
- 2 classified as **equivalent** (no observable behavior change):
 - **M0004**: Debug-only assertion in `remove_front()`
 - **M0024**: Formatting-only display branch in `Contig`
- 3 classified as **killable** — reveal genuine test gaps:
 - Missing boundary-condition tests for hole sizing
 - Exact-boundary segment assembly not exercised

Adjusted Mutation Adequacy

$$\text{Adequacy} = \frac{229}{333 - 2} = \frac{229}{331} = \mathbf{69.18\%}$$

Input Space Partitioning

Target: UDP parsing (wire/udp.rs)

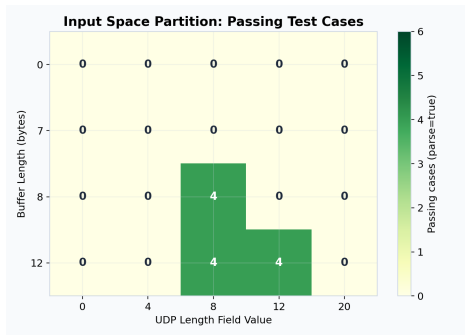
6 input dimensions:

1. Buffer length: {0, 7, 8, 12}
2. Length field: {0, 4, 8, 12, 20}
3. Destination port: {0, 53}
4. Checksum mode: {Valid, Zero, Invalid}
5. IP family: {IPv4, IPv6}
6. RX checking: {true, false}

$4 \times 5 \times 2 \times 3 \times 2 \times 2 = \mathbf{480}$ test frames

100% pass rate on both platforms

+216 IPv4 combinatorial test cases (all passing)



| Input Space Partitioning — Key Findings

Successful parsing requires **all** of:

- Buffer length ≥ 8 bytes
- Length field ≥ 8 and $\leq$ buffer length
- Non-zero destination port
- Valid or zero checksum (when checking enabled)

Notable finding

smoltcp accepts checksum=0 for UDP over IPv6 — a documented deviation from strict RFC 2460 expectations.

Security Fuzzing

Tool: cargo-fuzz + LLVM libFuzzer

5 fuzz targets:

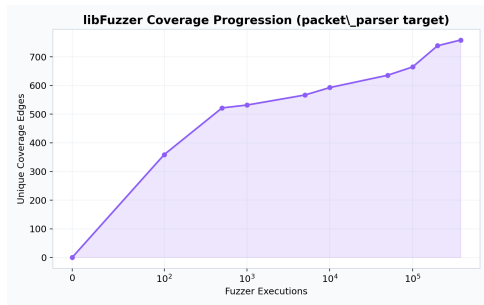
- packet_parser
- tcp_headers
- dhcp_header
- ieee802154_header
- sixlowpan_packet

Campaign: 180s per target

Executions: ~380K

Edges discovered: 759

✓ **Zero crash artifacts**



Fuzzing — Platform Portability Issue

Windows Fuzzing Blocked

All fuzzing attempts failed on Windows due to missing ASAN runtime DLLs and unresolved sanitizer-coverage symbols (MSVC/libFuzzer incompatibility).

Why zero crashes?

- Rust's bounds checking prevents buffer overflows
- No null pointer dereferences possible
- Safe integer arithmetic by default
- These are the exact vulnerability classes fuzzing typically finds in C/C++ stacks

RFC Conformance Testing

Case	RFC	Expected	Observed
IPv4 bad version	791	Reject	Reject
IPv4 IHL < 20B	791	Reject	Accept
IPv4 total < header	791	Reject	Reject
IPv4 bad checksum	791	Reject	Reject
IPv4 cksum disabled	config	Accept	Accept
UDP len < 8	768	Reject	Reject
UDP dst port = 0	smoltcp	Reject	Reject
UDP/IPv4 cksum = 0	768	Accept	Accept
UDP/IPv6 cksum = 0	2460	Reject	Accept

△ **Two specification deviations** identified (highlighted in orange)

RFC Deviations — Details

1. IPv4 header length < 20 bytes accepted

When IHL encodes a header < 20 bytes (RFC minimum), smoltcp accepts if the buffer is long enough.

- Deliberate design choice for performance
- Risk: crafted packets could bypass validation

2. UDP over IPv6 accepts checksum $= 0$

RFC 2460 mandates non-zero checksums for UDP/IPv6, but smoltcp accepts zero-checksum packets.

- Risk: corrupted datagrams may go undetected on IPv6 networks

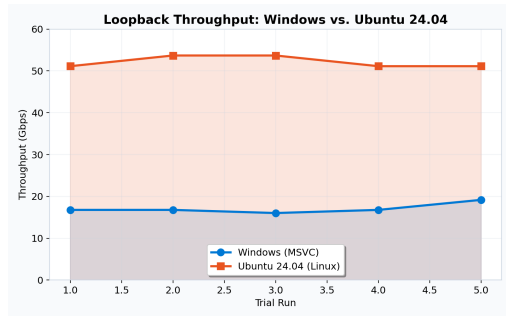
Performance Testing

Loopback throughput benchmark

- 128 MB data through internal stack
- Custom in-memory loopback harness
- 5 trial runs per platform

Results

- Ubuntu: **52.15 Gbps** avg
- Windows: **17.11 Gbps** avg
- **~3× difference**

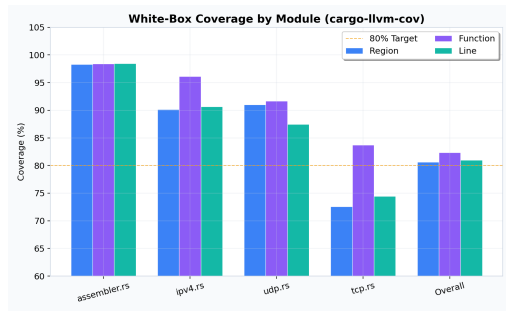


White-Box Test Data Adequacy

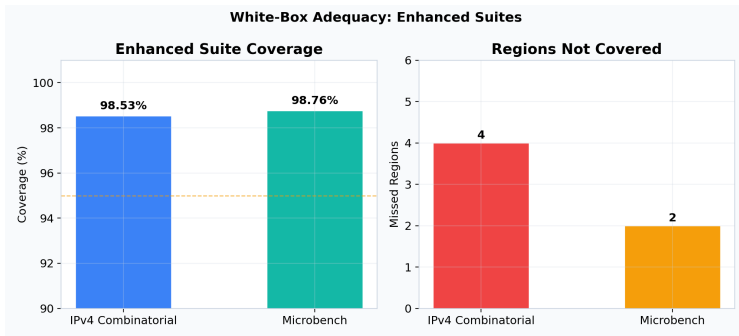
Tool: cargo-llvm-cov v0.8.5

Key metrics:

- assembler.rs: **98.28%** regions
- Protocol wire modules: 90+% parsing
- tcp.rs: 72.58% (lowest)
- IPv4 combinatorial: **98.53%**
- Microbenchmark: **98.76%**



Coverage with LCOV Artifacts



- Only **4 and 2 missed regions** respectively
- Remaining gaps are small but made **explicit and actionable**
- Provides concrete roadmap for future test additions

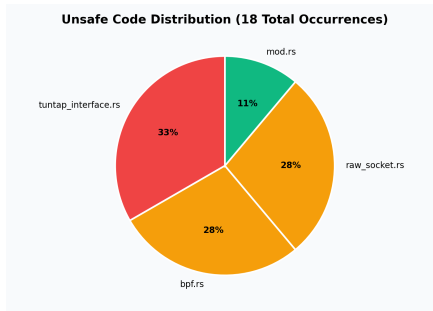
Software Safety

Panic Safety Testing

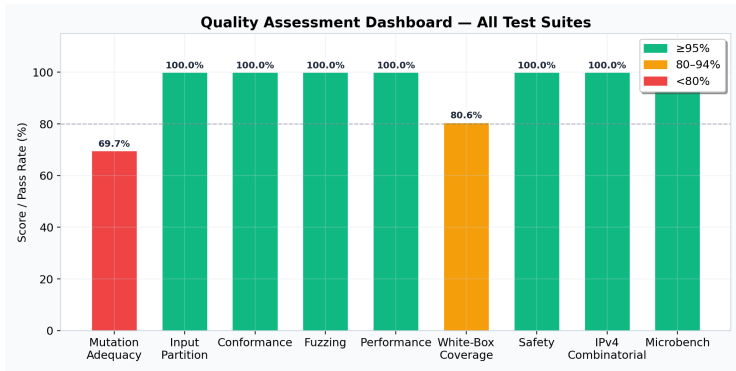
- Zero-length buffers
- Truncated headers
- Maximum-value fields
- All tests passed on both platforms

Unsafe Code Inventory

- 18 unsafe occurrences across 4 files
- All confined to FFI / PHY layer code
- Zero unsafe in core protocol parsing



Quality Assessment Dashboard



- Mutation adequacy (69.18%) — primary area for improvement
- All other dimensions achieved **80%+ pass rates**
- Core parsing and protocol handling is **robust**

Portability: Cross-Target Matrix

		Full	IPv4	IPv6	Dual	42 / 42 PASS	
		6 ISAs, 2 endiannesses hosted + bare-metal + netsim					
Hosted cross test	x86-64 glibc	✓	✓	✓	✓		
	x86 glibc / 32-bit	✓	✓	✓	✓		
	ARM64 glibc	✓	✓	✓	✓		
	ARMv7 hard-float	✓	✓	✓	✓		
	x86-64 musl	✓	✓	✓	✓		
	PowerPC 32 big-endian	✓	✓	✓	✓		
	MIPS32 little-endian	✓	✓	✓	✓		
	RISC-V 64 glibc	✓	✓	✓	✓		
Bare-metal cross check	ARMv7-M bare-metal / no_std	Min IPv4	Min IPv6	Alloc	N/A		
		✓	✓	✓	N/A		
Netsim integration	Release test 7 QEMU targets	x64	x86	arm64	armv7	ppc	mips
		✓	✓	✓	✓	✓	✓

- **42 runs** via cross: 8 hosted targets × 4 profiles
- Includes **big-endian PowerPC, 32-bit**, musl, and MIPS
- Netsim integration also passes (QEMU is slower off x86)
- Windows fuzzing blocked (MSVC/libFuzzer compatibility)

Concurrency: Loom Model Checking

3 loom checks, all PASS

6 / 6 orderings

0 panics / 0 deadlocks

Every model exhaustively covers every reachable thread ordering.

SCENARIO	THREADS	INTERLEAVINGS	RUNTIME
✓ Register vs. wake race waker_register_wake_race	2	2 orderings	~3 ms
✓ Concurrent replacement concurrent_waker_replacement	2	2 orderings	~3 ms
✓ Wake then re-register wake_then_reregister	2	2 orderings	~3 ms

Exhaustive across every reachable ordering

LOOM_MAX_PREEMPTIONS = 3

- Exhaustively explores async register/wake interleavings
- **3 loom checks, 6/6 orderings, 0 panics / 0 deadlocks**
- Loom also corrected a test-plan assertion (valid final state with no waker)

Conclusions & Key Takeaways

- ✓ **69.18% mutation adequacy** (adjusted) across 333 mutants
- ✓ **100% pass rate** on 696 input partition test cases
- ✓ **Zero crash artifacts** from coverage-guided fuzzing
- ✓ **Two RFC deviations** documented (IPv4 IHL, IPv6 UDP cksum)
- ✓ **3× performance gap** (Ubuntu vs. Windows)
- ✓ **No unsafe code** in core protocol modules
- ✓ **Cross-target & loom checks** completed with zero failures

smoltcp v0.12.0 is well-engineered with strong safety properties

Test suite would benefit from boundary-condition tests on wire protocol modules

Thank You

smoltcp v0.12.0 | Quality Assessment & Improvement